

# Visualization Literacy Analysis

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## Read in data files

```
#grab files from google drive (only have to do this once)
# source("getFromGoogleDrive.R")

#get the first 6? characters of each data file
#get unique values of these
# this is list of subject ids

raw_file_names <- list.files("../AccData")
first_six <- substr(raw_file_names, 1, 6)
sub_ids <- unique(first_six)

# fast_RT = data.frame(ParticipantId = character(),
#                       TrialName = character(),
#                       type = character(),
#                       time = numeric()
# )

all_data <- NULL

for (i in 1:length(sub_ids)){

  if (grepl("~$", sub_ids[i], fixed = T)){
    next
  }

  temp_file1 <- read_xlsx(paste0("../AccData/", sub_ids[i], "_1.xlsx")) %>%
    slice(1:17) %>%
    select(-starts_with("Order")) %>%
    rename(correct = 8)

  temp_file2 <- read_xlsx(paste0("../AccData/", sub_ids[i], "_2.xlsx")) %>%
    slice(1:17) %>%
    select(-starts_with("Order")) %>%
    rename(correct = 8)

  new_temp <- temp_file1 %>%
    bind_cols(temp_file2$correct) %>%
    rename(Correct_1 = correct, Correct_2 = "...9") %>%
    mutate(AnswerRT = TimeToBeginInput - TimeToReadQuestion)
```

```

all_data <- all_data %>%
  bind_rows(new_temp)
}

all_data <- all_data %>%
  rename(readRT = TimeToReadQuestion, totalRT = TimeToBeginInput)

#read in trialtype key (I created this from an early version of the previous paper)
trial_type_key <- read.csv("../trial_type_key.csv", stringsAsFactors = F)

all_data <- all_data %>%
  mutate(TrialType = trial_type_key$TrialType[match(TrialName, trial_type_key$TrialName)]) %>%
  mutate(TrialType = paste0("Type", TrialType))

```

## Basic checks

```

#how many participants per condition
all_data %>%
  group_by(ParticipantId, Condition) %>%
  summarize(ntrials = n()) %>%
  group_by(Condition) %>%
  summarize(nsubs = n()) %>%
  kable()

```

## `summarise()` has grouped output by 'ParticipantId'. You can override using the `.groups` argument.

| Condition         | nsubs |
|-------------------|-------|
| VR                | 50    |
| VR Monitor        | 39    |
| VR Monitor Stereo | 33    |

*#why is the balance so off?*

## Remove outliers based on RT

```

#removing on trial-by-trial basis

#remove answerRTs below 2000ms first
all_data_remove <- all_data %>%
  filter(AnswerRT >= 2000)
dim(all_data)[1]

```

```
## [1] 2074
```

```
dim(all_data_remove)[1]
```

```
## [1] 2067
```

```
#drops 7 trials
```

```
rt_data_summary <- all_data %>%  
  group_by(TrialName) %>%  
  summarize(meanAnswerRT = mean(AnswerRT, na.rm = T),  
            sdAnswerRT = sd(AnswerRT, na.rm = T),  
            UB = meanAnswerRT + 3*sdAnswerRT,  
            LB = meanAnswerRT - 3*sdAnswerRT)  
rt_data_summary
```

```
## # A tibble: 17 x 5  
##   TrialName      meanAnswerRT sdAnswerRT      UB      LB  
##   <chr>          <dbl>      <dbl>    <dbl>  <dbl>  
## 1 BarChartQ1      27611.    14762.   71897. -16675.  
## 2 BarChartQ2      16921.    13338.   56935. -23093.  
## 3 BarChartQ3      15609.     9948.  45452. -14235.  
## 4 BarChartQ4      10288.     8978.  37222. -16646.  
## 5 LineChartQ1      49185.   29505. 137699. -39329.  
## 6 LineChartQ2      40127.   31660. 135107. -54854.  
## 7 LineChartQ3      27190.   17504.  79701. -25322.  
## 8 LineChartQ4      14779.   16568.  64483. -34925.  
## 9 LineChartQ5      27877.   17250.  79628. -23874.  
## 10 ScatterplotQ1    32801.   24294. 105682. -40080.  
## 11 ScatterplotQ2    29636.   23980. 101576. -42304.  
## 12 ScatterplotQ3    45580.   33014. 144623. -53463.  
## 13 ScatterplotQ4    35987.   25402. 112194. -40219.  
## 14 ScatterplotQ5    59805.   42684. 187859. -68248.  
## 15 SurfacePlotQ1    56608.   36042. 164733. -51516.  
## 16 SurfacePlotQ2    65110.   50062. 215297. -85076.  
## 17 SurfacePlotQ3    48373.   37424. 160646. -63900.
```

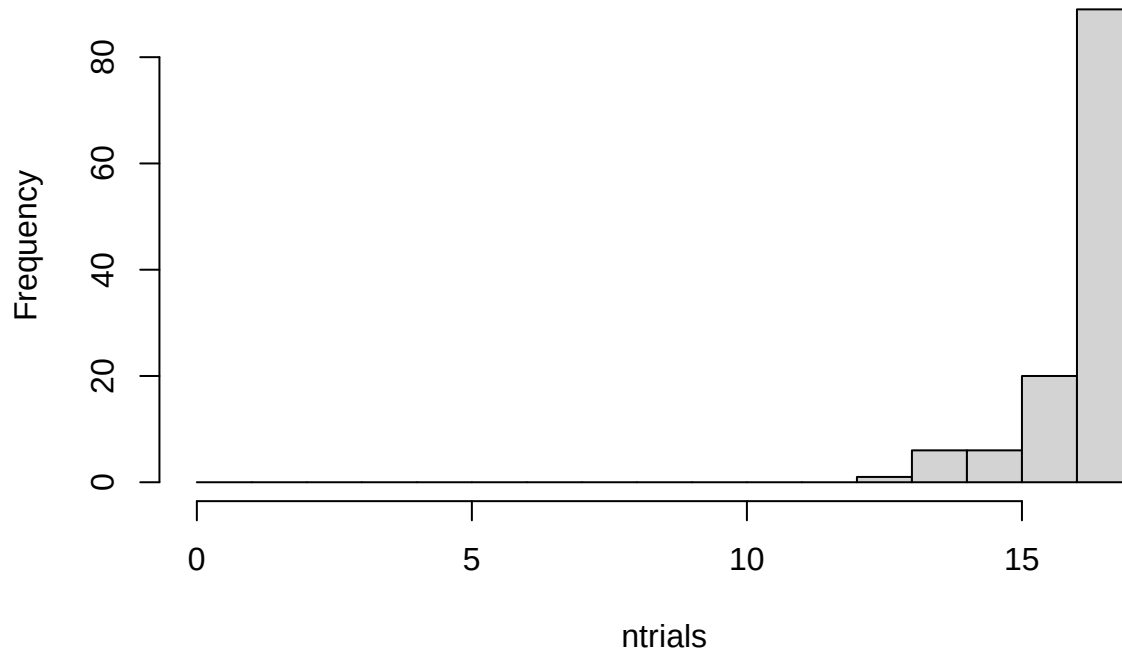
```
all_data_no_outliers <- all_data_remove %>%  
  group_by(TrialName) %>%  
  filter(!(abs(AnswerRT - mean(AnswerRT)) > 3.0*sd(AnswerRT)))  
dim(all_data_no_outliers)[1]
```

```
## [1] 2020
```

```
#drops 47 more trials
```

```
all_data_no_outliers %>%  
  group_by(ParticipantId) %>%  
  summarize(ntrials = n()) %>%  
  with(hist(ntrials, breaks = 0:17))
```

## Histogram of ntrials



*##maybe consider replacing outliers with means instead of removing them?*

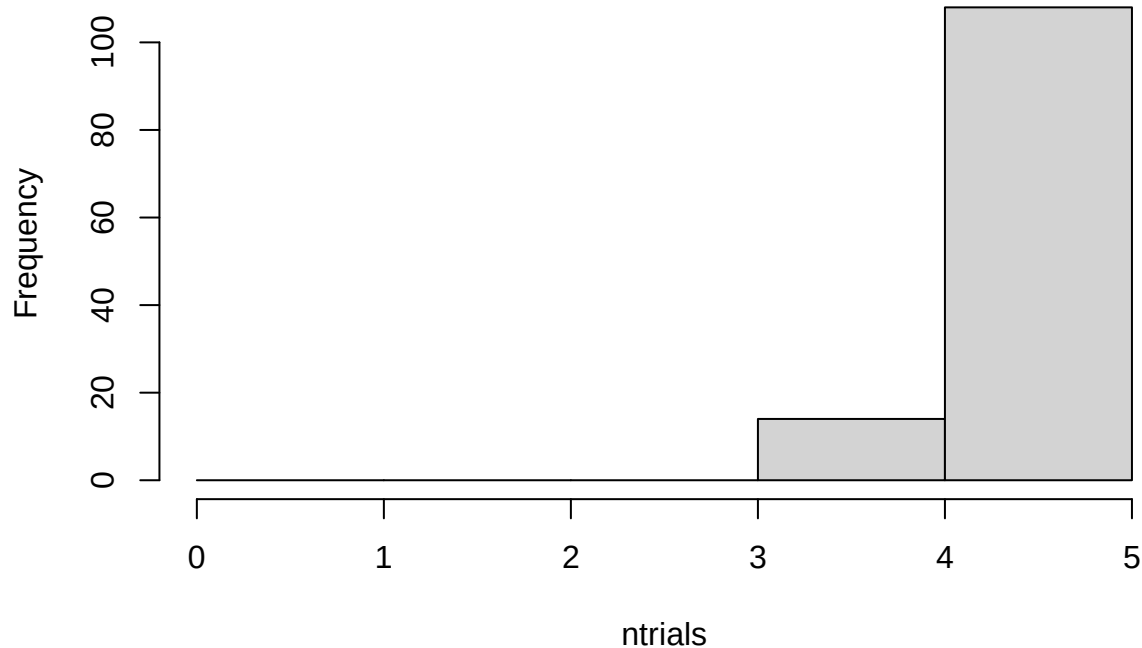
### Extract Only One Type of Chart

Create a dataframe from the data with no outliers that contains only XChartQ1 through XChartQ4 trials for the following analysis.

```
all_data_no_outliers <- all_data_no_outliers %>%  
  filter(grepl("LineChartQ[0-9]", TrialName))
```

```
all_data_no_outliers %>%  
  group_by(ParticipantId) %>%  
  summarize(ntrials = n()) %>%  
  with(hist(ntrials, breaks = 0:5))
```

## Histogram of ntrials



Compare RTs for conditions and question type (three types: identify, relate, predict)

```
#compare read times (should be no differences of condition)
#compare answer times (potentially a difference)
```

```
trial_type_means <- all_data_no_outliers %>%
  group_by(ParticipantId, Condition, TrialType) %>%
  summarize(mean_readRT = mean(readRT),
            mean_answerRT = mean(AnswerRT),
            n = n())
```

```
## `summarise()` has grouped output by 'ParticipantId', 'Condition'. You can override using
## the `.groups` argument.
```

```
# first, make .csv files in wide format to double check in statview
```

```
read_rt_type_wider <- trial_type_means %>%
  select(ParticipantId, Condition, TrialType, mean_readRT) %>%
  pivot_wider(names_from = TrialType, values_from = mean_readRT)
```

```
# fill in na values with mean for type1, type2 and type3 RT
```

```
# the following is supposed to work, but we get NaN from calculation of mean here, not sure why...
```

```
#read_rt_type_wider_clean <- read_rt_type_wider %>% mutate(across(Type1, ~replace_na(., mean(., na.rm=T))
```

```
read_rt_type_wider_clean <- read_rt_type_wider %>% replace_na(list(Type1 = mean(read_rt_type_wider$Type1, na.rm=T))
```

```
read_rt_type_wider_clean <- read_rt_type_wider_clean %>% replace_na(list(Type2 = mean(read_rt_type_wider$Type2, na.rm=T))
```

```

read_rt_type_wider_clean <- read_rt_type_wider_clean %>% replace_na(list(Type3 = mean(read_rt_type_wider_clean$Type3)))

answer_rt_type_wider <- trial_type_means %>%
  select(ParticipantId, Condition, TrialType, mean_answerRT) %>%
  pivot_wider(names_from = TrialType, values_from = mean_answerRT)
write.csv(read_rt_type_wider, file = "readtypeRTs.csv", row.names = F)
write.csv(answer_rt_type_wider, file = "answertypeRTs.csv", row.names = F)

# fill in na values with mean for type1, type2 and type3 RT
answer_rt_type_wider_clean <- answer_rt_type_wider %>% replace_na(list(Type1 = mean(answer_rt_type_wider_clean$Type1),
  Type2 = mean(answer_rt_type_wider_clean$Type2),
  Type3 = mean(answer_rt_type_wider_clean$Type3)))

#### READ RTs ####

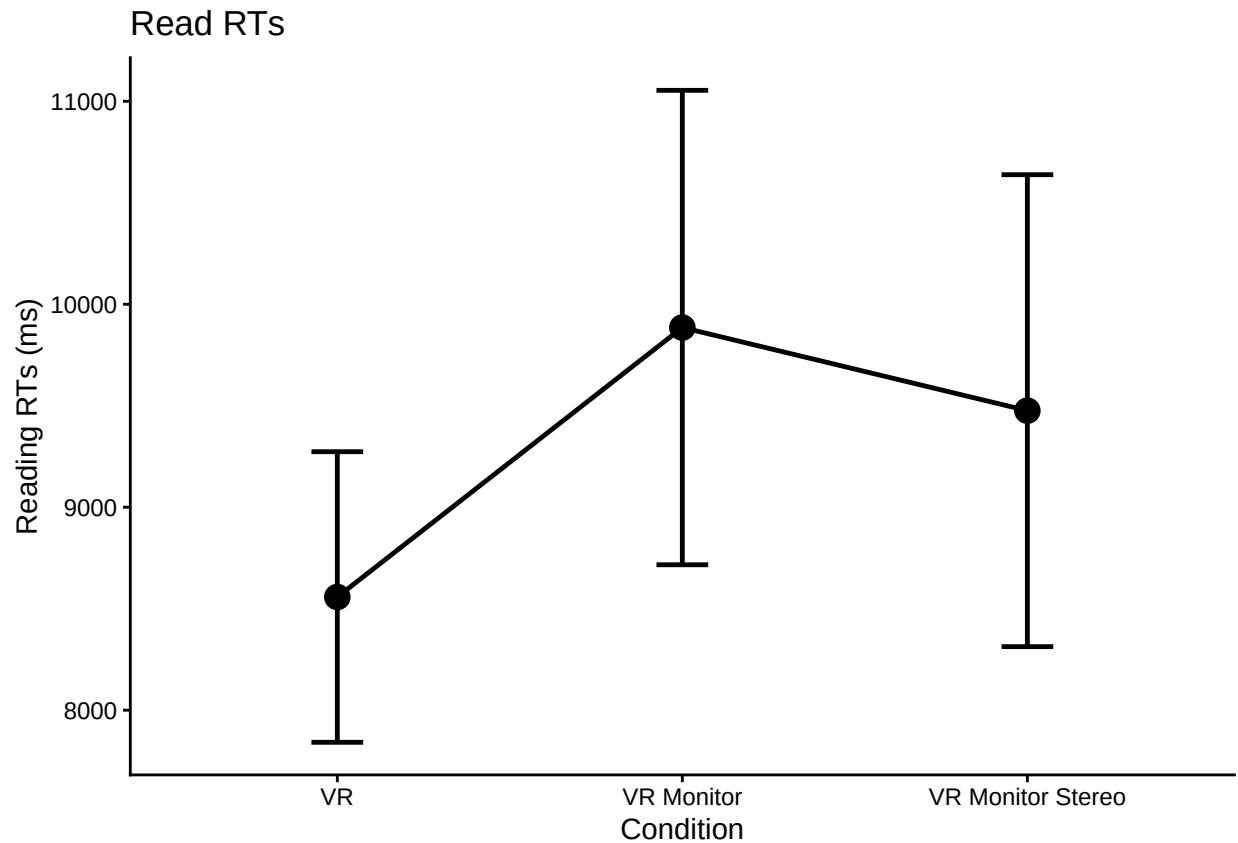
#make some plots

#just condition main effect
readplot1 <- all_data_no_outliers %>%
  group_by(ParticipantId, Condition) %>%
  summarize(overallmean = mean(readRT)) %>%
  group_by(Condition) %>%
  summarize(overall_condition_mean = mean(overallmean),
    se = std.error(overallmean),
    n = n(),
    CI = qt(0.975,df=n-1)*se)

## `summarise()` has grouped output by 'ParticipantId'. You can override using the `.groups`
## argument.

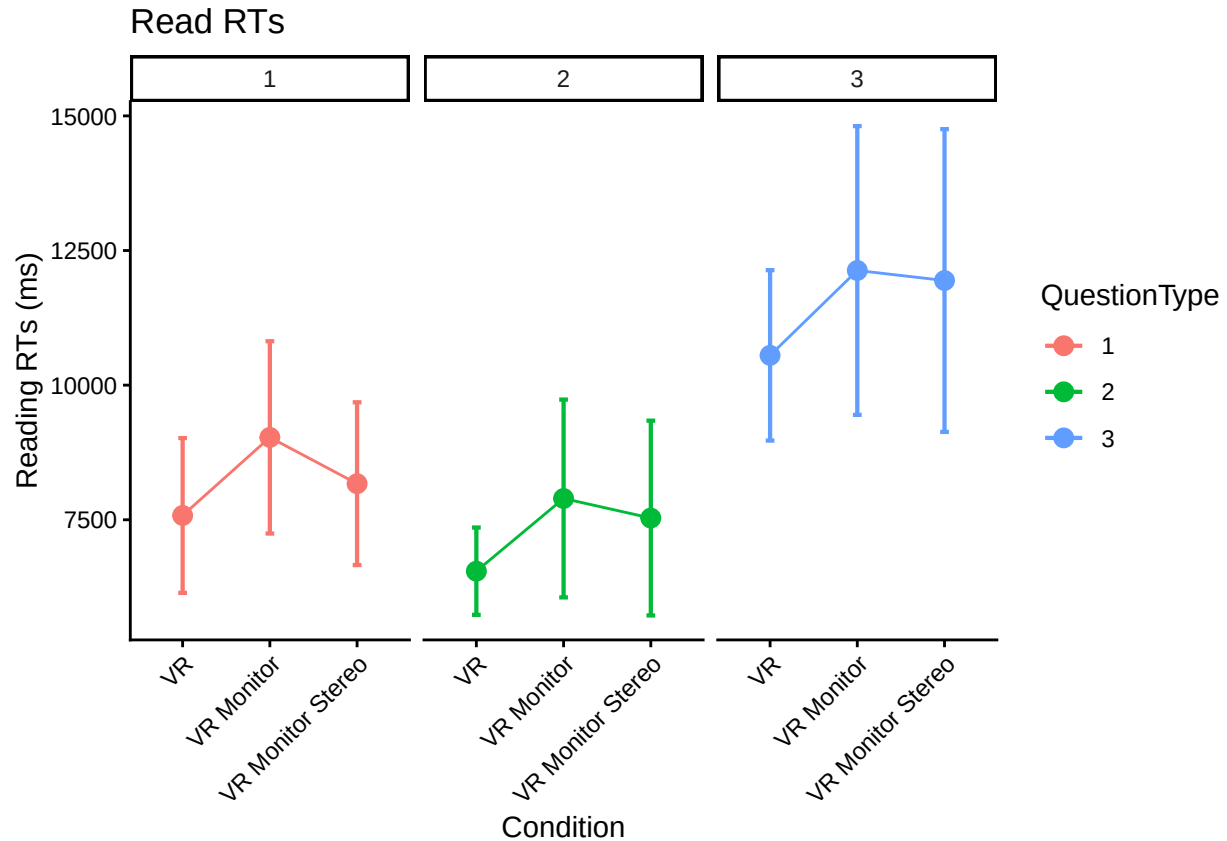
ggplot(readplot1, aes(Condition,
  overall_condition_mean,
  group = 1,
  ymin = overall_condition_mean - CI,
  ymax = overall_condition_mean + CI)) +
  theme_classic() +
  geom_point(size = 4) +
  geom_errorbar(width = .15, linewidth = 0.85) +
  geom_line(linewidth = 0.85) +
  labs(y = "Reading RTs (ms)", title = "Read RTs")

```



```
#make a little plot using wide format
superbPlot(read_rt_type_wider_clean,
  BSFactors = "Condition",
  WSFactors = "QuestionType(3)",
  variables = c("Type1", "Type2", "Type3"),
  statistic = "meanNArm",
  errorbar = "CI",
  gamma = 0.95,
  adjustments = list(
    purpose = "difference"
  ),
  plotLayout = "line",
  factorOrder = c("Condition", "QuestionType")
) +
theme_classic() +
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust=1))+
facet_wrap(vars(QuestionType))+
labs(y = "Reading RTs (ms)", title = "Read RTs")
```

## superb::ADVICE: Some of the groups' variances are heterogeneous. Consider using purpose="tryon".



```
# there are missing values in anova for some participants because they don't
# do all trial types necessarily. Probably a better way to do this, but
# we widen table to make na easier to identify, fill in na with mean of column,
# then make back to narrow version for the anova here
x <- trial_type_means %>%
  select(ParticipantId, Condition, TrialType, mean_readRT) %>%
  pivot_wider(names_from = TrialType, values_from = mean_readRT)
x_clean <- x
x_clean$Type1[is.na(x$Type1)] <- mean(x$Type1, na.rm=TRUE)
x_clean$Type2[is.na(x$Type2)] <- mean(x$Type2, na.rm=TRUE)
x_clean$Type3[is.na(x$Type3)] <- mean(x$Type3, na.rm=TRUE)
readRT_means_clean <- x_clean %>%
  pivot_longer(Type1:Type3, names_to="TrialType", values_to = "mean_readRT")

read_rt_type_anova <- aov_ez(id = "ParticipantId",
  dv = "mean_readRT",
  data = readRT_means_clean,
  within = "TrialType",
  between = "Condition",
  anova_table = list(es = "pes") #might want to double-check these
)

## Converting to factor: Condition
## Contrasts set to contr.sum for the following variables: Condition
kable(nice(read_rt_type_anova))
```



| Effect              | df           | MSE         | F         | pes  | p.value |
|---------------------|--------------|-------------|-----------|------|---------|
| Condition           | 2, 119       | 26360029.53 | 2.82 +    | .045 | .064    |
| TrialType           | 1.80, 214.66 | 12333808.46 | 52.26 *** | .305 | <.001   |
| Condition:TrialType | 3.61, 214.66 | 12333808.46 | 0.16      | .003 | .945    |

```
#posthoc test for trial type
```

```
pairs(emmeans(read_rt_type_anova, "TrialType"), adjust = "Tukey")
```

```
## contrast estimate SE df t.ratio p.value
```

```
## Type1 - Type2 936 366 119 2.554 0.0317
```

```
## Type1 - Type3 -3282 491 119 -6.677 <.0001
```

```
## Type2 - Type3 -4218 433 119 -9.744 <.0001
```

```
##
```

```
## Results are averaged over the levels of: Condition
```

```
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
#posthoc test for condition, still using Tukey
```

```
pairs(emmeans(read_rt_type_anova, "Condition"), adjust = "Tukey")
```

```
## contrast estimate SE df t.ratio p.value
```

```
## VR - VR Monitor -1459 633 119 -2.304 0.0591
```

```
## VR - VR Monitor Stereo -989 665 119 -1.488 0.3004
```

```
## VR Monitor - VR Monitor Stereo 470 701 119 0.670 0.7813
```

```
##
```

```
## Results are averaged over the levels of: TrialType
```

```
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
#posthoc test for condition, others seem to be scheffe, sidak, bonferroni, dunnett, mvt, none
```

```
pairs(emmeans(read_rt_type_anova, "Condition"), adjust = "scheffe")
```

```
## contrast estimate SE df t.ratio p.value
```

```
## VR - VR Monitor -1459 633 119 -2.304 0.0746
```

```
## VR - VR Monitor Stereo -989 665 119 -1.488 0.3340
```

```
## VR Monitor - VR Monitor Stereo 470 701 119 0.670 0.7993
```

```
##
```

```
## Results are averaged over the levels of: TrialType
```

```
## P value adjustment: scheffe method with rank 2
```

```
#### ANSWER RTs ####
```

```
#make some plots
```

```
#just condition main effect
```

```
answerplot1 <- all_data_no_outliers %>%
```

```
  group_by(ParticipantId, Condition) %>%
```

```
  summarize(overallmean = mean(AnswerRT)) %>%
```

```
  group_by(Condition) %>%
```

```
  summarize(overall_condition_mean = mean(overallmean),
```

```
            se = std.error(overallmean),
```

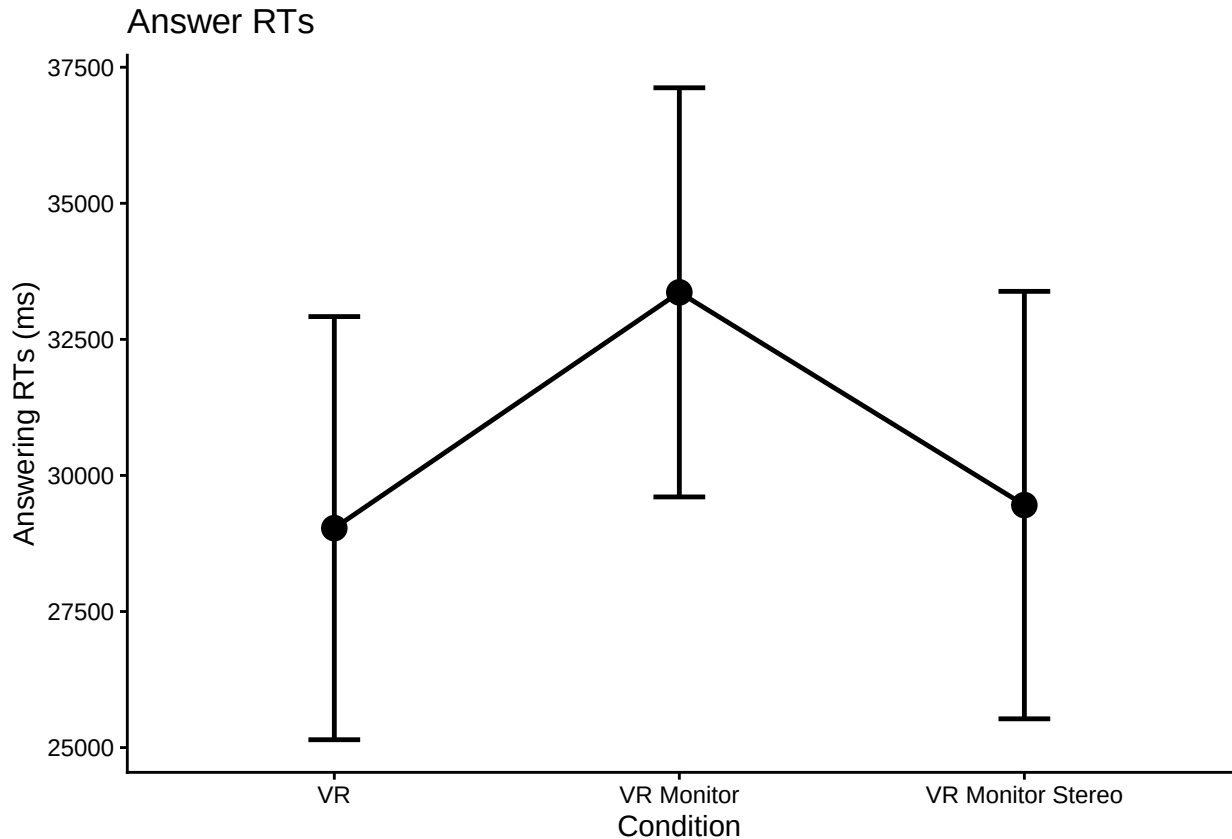
```
            n = n(),
```

```
            CI = qt(0.975,df=n-1)*se)
```

```
## `summarise()` has grouped output by 'ParticipantId'. You can override using the `.groups`
```

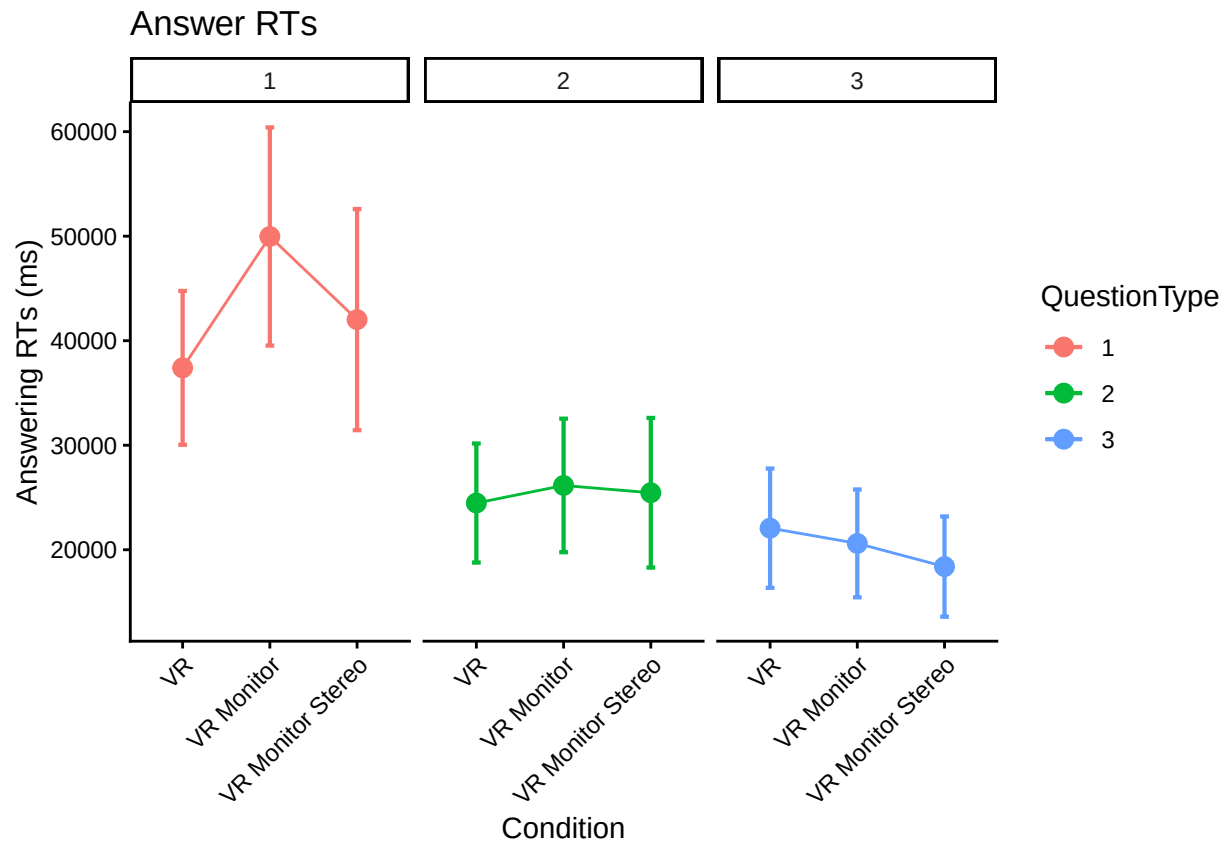
```
## argument.
```

```
ggplot(answerplot1, aes(Condition,
                        overall_condition_mean,
                        group = 1,
                        ymin = overall_condition_mean - CI,
                        ymax = overall_condition_mean + CI)) +
  theme_classic() +
  geom_point(size = 4) +
  geom_errorbar(width = .15, linewidth = 0.85) +
  geom_line(linewidth = 0.85) +
  labs(y = "Answering RTs (ms)", title = "Answer RTs")
```



```
#make the little plot
superbPlot(answer_rt_type_wider_clean,
  BSFactors = "Condition",
  WSFactors = "QuestionType(3)",
  variables = c("Type1", "Type2", "Type3"),
  statistic = "meanNArm",
  errorbar = "CI",
  gamma = 0.95,
  adjustments = list(
    purpose = "difference"
  ),
  plotLayout = "line",
  factorOrder = c("Condition", "QuestionType")
) +
theme_classic() +
```

```
theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust=1))+
facet_wrap(vars(QuestionType))+
labs(y = "Answering RTs (ms)", title = "Answer RTs")
```



```
# there are missing values in anova for some participants because they don't
# do all trial types necessarily. Probably a better way to do this, but
# we widen table to make na easier to identify, fill in na with mean of column,
# then make back to narrow version for the anova here
x <- trial_type_means %>%
  select(ParticipantId, Condition, TrialType, mean_answerRT) %>%
  pivot_wider(names_from = TrialType, values_from = mean_answerRT)
x_clean <- x
x_clean$Type1[is.na(x$Type1)] <- mean(x$Type1, na.rm=TRUE)
x_clean$Type2[is.na(x$Type2)] <- mean(x$Type2, na.rm=TRUE)
x_clean$Type3[is.na(x$Type3)] <- mean(x$Type3, na.rm=TRUE)
answerRT_means_clean <- x_clean %>%
  pivot_longer(Type1:Type3, names_to="TrialType", values_to = "mean_answerRT")

answer_rt_type_anova <- aov_ez(id = "ParticipantId",
  dv = "mean_answerRT",
  data = answerRT_means_clean,
  within = "TrialType",
  between = "Condition",
  anova_table = list(es = "pes") #might want to double-check these
)
```

```
## Converting to factor: Condition
## Contrasts set to contr.sum for the following variables: Condition
kable(nice(answer_rt_type_anova))
```

| Effect              | df           | MSE          | F         | pes  | p.value |
|---------------------|--------------|--------------|-----------|------|---------|
| Condition           | 2, 119       | 406094394.67 | 1.61      | .026 | .205    |
| TrialType           | 1.74, 207.12 | 209492293.80 | 93.13 *** | .439 | <.001   |
| Condition:TrialType | 3.48, 207.12 | 209492293.80 | 3.44 *    | .055 | .013    |

```
#posthoc test for trial type
pairs(emmeans(answer_rt_type_anova, "TrialType"), adjust = "Tukey")

## contrast      estimate    SE  df t.ratio p.value
## Type1 - Type2    17771 1930 119   9.185 <.0001
## Type1 - Type3    22776 1900 119  12.012 <.0001
## Type2 - Type3     5004 1380 119   3.638 0.0012
##
## Results are averaged over the levels of: Condition
## P value adjustment: tukey method for comparing a family of 3 estimates
#Question Type 3 much faster than Type 1/Type 2 (this is answering times)
```

```
#posthoc test for condition, still using Tukey
pairs(emmeans(answer_rt_type_anova, "Condition"), adjust = "Tukey")

## contrast              estimate    SE  df t.ratio p.value
## VR - VR Monitor        -4263 2490 119  -1.715 0.2037
## VR - VR Monitor Stereo  -641 2610 119  -0.246 0.9673
## VR Monitor - VR Monitor Stereo  3623 2750 119   1.316 0.3889
##
## Results are averaged over the levels of: TrialType
## P value adjustment: tukey method for comparing a family of 3 estimates
ref <- emmeans(answer_rt_type_anova, ~Condition|TrialType)
pairs(ref, adjust = "Tukey")
```

```
## TrialType = Type1:
## contrast              estimate    SE  df t.ratio p.value
## VR - VR Monitor        -12569 4400 119  -2.860 0.0138
## VR - VR Monitor Stereo  -4616 4610 119  -1.000 0.5781
## VR Monitor - VR Monitor Stereo  7953 4870 119   1.634 0.2353
##
## TrialType = Type2:
## contrast              estimate    SE  df t.ratio p.value
## VR - VR Monitor        -1681 3020 119  -0.557 0.8430
## VR - VR Monitor Stereo  -982 3170 119  -0.310 0.9484
## VR Monitor - VR Monitor Stereo  699 3340 119   0.209 0.9762
##
## TrialType = Type3:
## contrast              estimate    SE  df t.ratio p.value
## VR - VR Monitor        1459 2600 119   0.561 0.8408
## VR - VR Monitor Stereo  3675 2730 119   1.347 0.3722
## VR Monitor - VR Monitor Stereo  2216 2880 119   0.770 0.7221
```

```
##
## P value adjustment: tukey method for comparing a family of 3 estimates
#plot and interaction suggests that the conditions have different effects for different
#question types. posthoc tests indicate a difference between VR and VRMonitor for QType 1
```

## Accuracy

```
# do interrater reliability: Fits each trial as independent with two raters
# That is, this ignores participants (complete pooling, which may be problematic)
# 1410 trials from N = 85 participants, not exactly 17 trials per participant
# b/c of excluded trials
irr::icc(select(ungroup(all_data_no_outliers), Correct_1, Correct_2))
```

```
## Single Score Intraclass Correlation
##
## Model: oneway
## Type : consistency
##
## Subjects = 596
## Raters = 2
## ICC(1) = 0.975
##
## F-Test, H0: r0 = 0 ; H1: r0 > 0
## F(595,596) = 78.2 , p = 0
##
## 95%-Confidence Interval for ICC Population Values:
## 0.97 < ICC < 0.978
```

```
head(all_data_no_outliers)
```

```
## # A tibble: 6 x 11
## # Groups:   TrialName [5]
## ParticipantId Condition TrialNumber TrialName readRT totalRT UserInput Correct_1
## <chr> <chr> <dbl> <chr> <dbl> <dbl> <chr> <dbl>
## 1 AA0176 VR Monitor 1 LineChartQ1 16001. 70295. 1.0 0.5
## 2 AA0176 VR Monitor 2 LineChartQ2 14398. 98640. 6.0 1
## 3 AA0176 VR Monitor 3 LineChartQ3 9875. 52812. neighborhood 1 0
## 4 AA0176 VR Monitor 4 LineChartQ4 8780. 13538. Neighborhood 2 1
## 5 AA0176 VR Monitor 5 LineChartQ5 18755. 58351. neighborhood 2 0
## 6 AA0876 VR 1 LineChartQ1 37107. 86386. 2.0 0.5
## # i 3 more variables: Correct_2 <dbl>, AnswerRT <dbl>, TrialType <chr>
```

```
# account for non-independence?
#Data has to be narrow
#https://cran.r-project.org/web/packages/cccrm/index.html
icc.dat.narrow <- all_data_no_outliers %>%
  pivot_longer(cols = c('Correct_1', 'Correct_2'))
```

```
colnames(icc.dat.narrow)[10] <- "Rater"
icc.dat.narrow$Rater <- as.factor(icc.dat.narrow$Rater)
```

```
#Longitudinal concordance for two raters
#By all 17 trials (by items)
# icc.rm.trial.name <- ccclon(icc.dat.narrow,
```

```

#                                     "value", "ParticipantId", "TrialName", "Rater")
#   icc.rm.trial.name
#   summary(icc.rm.trial.name)

#By time using trial numbers (by time)
icc.rm.trial.number <- ccclon(icc.dat.narrow,
                             "value", "ParticipantId", "TrialNumber", "Rater" )

## Warning: `ccclon()` was deprecated in cccrm 3.0.0.
## i Please use `ccc_vc()` instead.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was generated.
icc.rm.trial.number

## CCC estimated by variance components:
##          CCC    LL CI 95%    UL CI 95%          SE CCC
## 0.973544416 0.968737371 0.977620716 0.002256246

summary(icc.rm.trial.number)

##          Subjects Subjects-Method   Subjects-Time          Method          Error
##    2.568737e-02    3.887417e-04    1.750894e-01    3.748606e-05    5.029780e-03
##
## CCC estimated by variance components
##          CCC    LL CI 95%    UL CI 95%          SE CCC
## 0.973544416 0.968737371 0.977620716 0.002256246

# A similar approach
#https://peerj.com/articles/9850/
# test.mod <-
#   lcc(data = hue, subject = "Fruit", resp = "H_mean",
#        method = "Method", time = "Time", qf = 2, qr = 1)
# summary(test.mod)

# icc.dat.narrow$ParticipantId <- as.factor(icc.dat.narrow$ParticipantId)
# icc.dat.narrow$Rater <- as.factor(icc.dat.narrow$Rater)
#
#
# icc.rm.another <-
#   lcc(data = icc.dat.narrow,
#        subject = "ParticipantId",
#        resp = "value",
#        method = "Rater",
#        time = "TrialNumber",
#        qf = 1, #polynomial trends (1 to 3)
#        qr = 0, #random effects (0 is random int only, default)
#        components = TRUE)
#
# #LCC = Longitudinal Concordance Correlation
# #LPC = Longitudinal Pearson Correlation
# #LA  = Longitudinal Accuracy
# icc.rm.another
#
# #Model coefficients near 1 but only the the last one has a correctly
# #estimated model. Maybe because it's more or less flat over TrialNumber?
# lccPlot(icc.rm.another, type = "lpc")

```

```

# lccPlot(icc.rm.another, type = "lcc")
# lccPlot(icc.rm.another, type = "la")

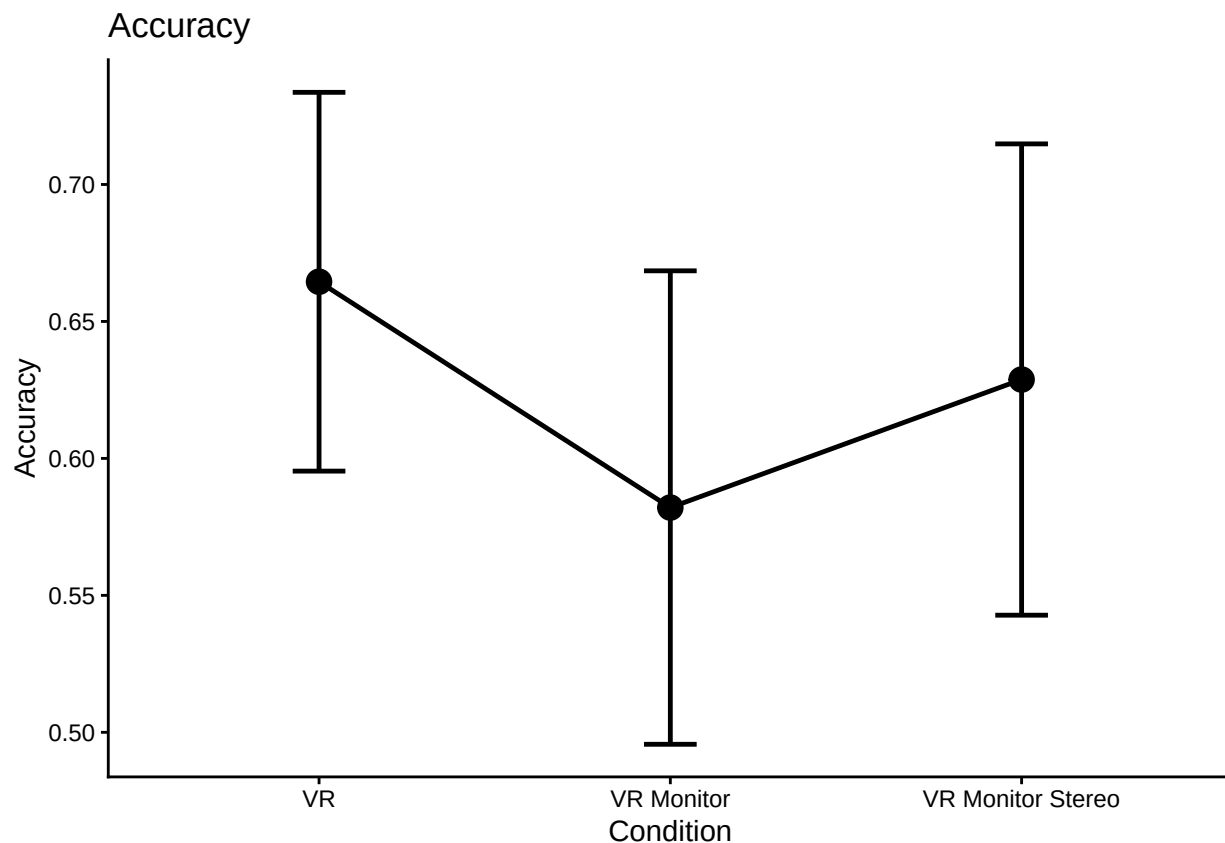
all_data_no_outliers <- all_data_no_outliers %>%
  rowwise() %>%
  mutate(Correct_Avg = mean(c(Correct_1, Correct_2)))

#just condition main effect
accplot1 <- all_data_no_outliers %>%
  group_by(ParticipantId, Condition) %>%
  summarize(overallmean = mean(Correct_Avg)) %>%
  group_by(Condition) %>%
  summarize(overall_condition_mean = mean(overallmean),
            se = std.error(overallmean),
            n = n(),
            CI = qt(0.975,df=n-1)*se)

## `summarise()` has grouped output by 'ParticipantId'. You can override using the `groups`
## argument.

ggplot(accplot1, aes(Condition,
                     overall_condition_mean,
                     group = 1,
                     ymin = overall_condition_mean - CI,
                     ymax = overall_condition_mean + CI)) +
  theme_classic() +
  geom_point(size = 4) +
  geom_errorbar(width = .15, linewidth = 0.85) +
  geom_line(linewidth = 0.85) +
  labs(y = "Accuracy", title = "Accuracy")

```



```
trial_type_mean_acc <- all_data_no_outliers %>%
  group_by(ParticipantId, Condition, TrialType) %>%
  summarize(mean_acc = mean(Correct_Avg),
            n = n())
```

## `summarise()` has grouped output by 'ParticipantId', 'Condition'. You can override using  
## the `.groups` argument.

```
acc_type_wider <- trial_type_mean_acc %>%
  select(ParticipantId, Condition, TrialType, mean_acc) %>%
  pivot_wider(names_from = TrialType, values_from = mean_acc)
write.csv(acc_type_wider, file = "typeacc.csv", row.names = F)
```

*# replace na with mean*

```
acc_type_wider_clean <- acc_type_wider %>% replace_na(list(Type1 = mean(acc_type_wider$Type1, na.rm=TRUE)))
acc_type_wider_clean <- acc_type_wider_clean %>% replace_na(list(Type2 = mean(acc_type_wider$Type2, na.rm=TRUE)))
acc_type_wider_clean <- acc_type_wider_clean %>% replace_na(list(Type3 = mean(acc_type_wider$Type3, na.rm=TRUE)))
```

*#make the little plot*

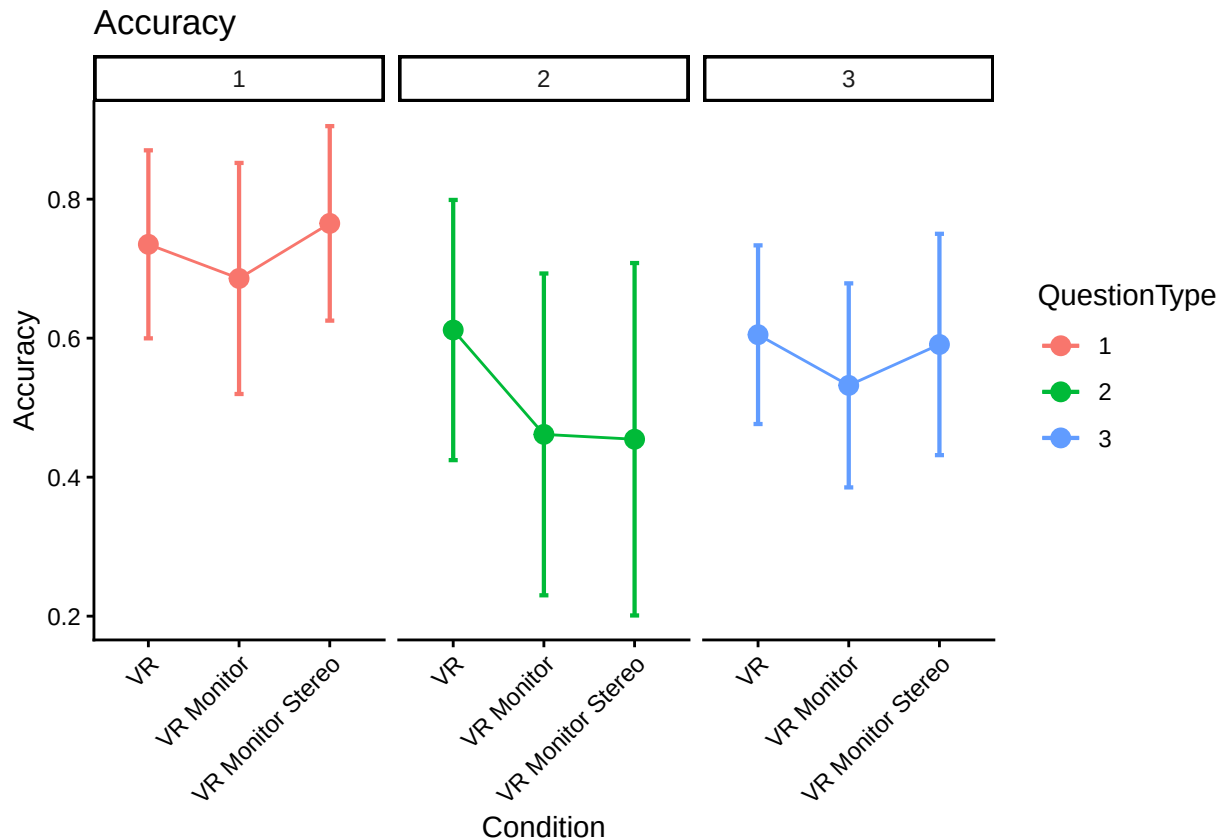
```
superbPlot(acc_type_wider_clean,
  BSFactors = "Condition",
  WSFactors = "QuestionType(3)",
  variables = c("Type1", "Type2", "Type3"),
  statistic = "meanNArm",
  errorbar = "CI",
  gamma = 0.95,
  adjustments = list(
```



```

    purpose      = "difference"
  ),
  plotLayout = "line",
  factorOrder = c("Condition", "QuestionType")
) +
theme_classic() +
theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust=1))+
facet_wrap(vars(QuestionType))+
labs(y = "Accuracy", title = "Accuracy")

```



```

# there are missing values in anova for some participants because they don't
# do all trial types necessarily. Probably a better way to do this, but
# we widen table to make na easier to identify, fill in na with mean of column,
# then make back to narrow version for the anova here
x <- trial_type_mean_acc %>%
  select(ParticipantId, Condition, TrialType, mean_acc) %>%
  pivot_wider(names_from = TrialType, values_from = mean_acc)
x_clean <- x
x_clean$Type1[is.na(x$Type1)] <- mean(x$Type1, na.rm=TRUE)
x_clean$Type2[is.na(x$Type2)] <- mean(x$Type2, na.rm=TRUE)
x_clean$Type3[is.na(x$Type3)] <- mean(x$Type3, na.rm=TRUE)
acc_means_clean <- x_clean %>%
  pivot_longer(Type1:Type3, names_to="TrialType", values_to = "mean_acc")

acc_type_anova <- aov_ez(id = "ParticipantId",
  dv = "mean_acc",

```

```

data = acc_means_clean,
within = "TrialType",
between = "Condition",
anova_table = list(es = "pes") #might want to double-check these
)

```

```

## Converting to factor: Condition
## Contrasts set to contr.sum for the following variables: Condition
kable(nice(acc_type_anova))

```

| Effect              | df           | MSE  | F         | pes  | p.value |
|---------------------|--------------|------|-----------|------|---------|
| Condition           | 2, 119       | 0.21 | 1.27      | .021 | .284    |
| TrialType           | 1.75, 208.00 | 0.14 | 12.65 *** | .096 | <.001   |
| Condition:TrialType | 3.50, 208.00 | 0.14 | 0.83      | .014 | .497    |

```

#posthoc test for trial type
pairs(emmeans(acc_type_anova, "TrialType"), adjust = "Tukey")

## contrast      estimate      SE  df t.ratio p.value
## Type1 - Type2   0.2194 0.0500 119   4.388  0.0001
## Type1 - Type3   0.1527 0.0354 119   4.310  0.0001
## Type2 - Type3  -0.0667 0.0474 119  -1.408  0.3400
##
## Results are averaged over the levels of: Condition
## P value adjustment: tukey method for comparing a family of 3 estimates
#Question Type 3 much faster than Type 1/Type 2 (this is answering times)

ref <- emmeans(acc_type_anova, ~Condition|TrialType)

pairs(ref, adjust = "Tukey")

```

```

## TrialType = Type1:
## contrast              estimate      SE  df t.ratio p.value
## VR - VR Monitor         0.04910 0.0707 119   0.694  0.7673
## VR - VR Monitor Stereo  -0.03015 0.0742 119  -0.406  0.9132
## VR Monitor - VR Monitor Stereo -0.07925 0.0783 119  -1.012  0.5706
##
## TrialType = Type2:
## contrast              estimate      SE  df t.ratio p.value
## VR - VR Monitor         0.15016 0.1050 119   1.436  0.3257
## VR - VR Monitor Stereo   0.15715 0.1100 119   1.432  0.3280
## VR Monitor - VR Monitor Stereo 0.00699 0.1160 119   0.060  0.9980
##
## TrialType = Type3:
## contrast              estimate      SE  df t.ratio p.value
## VR - VR Monitor         0.07295 0.0682 119   1.069  0.5350
## VR - VR Monitor Stereo   0.01409 0.0716 119   0.197  0.9789
## VR Monitor - VR Monitor Stereo -0.05886 0.0755 119  -0.779  0.7165
##
## P value adjustment: tukey method for comparing a family of 3 estimates

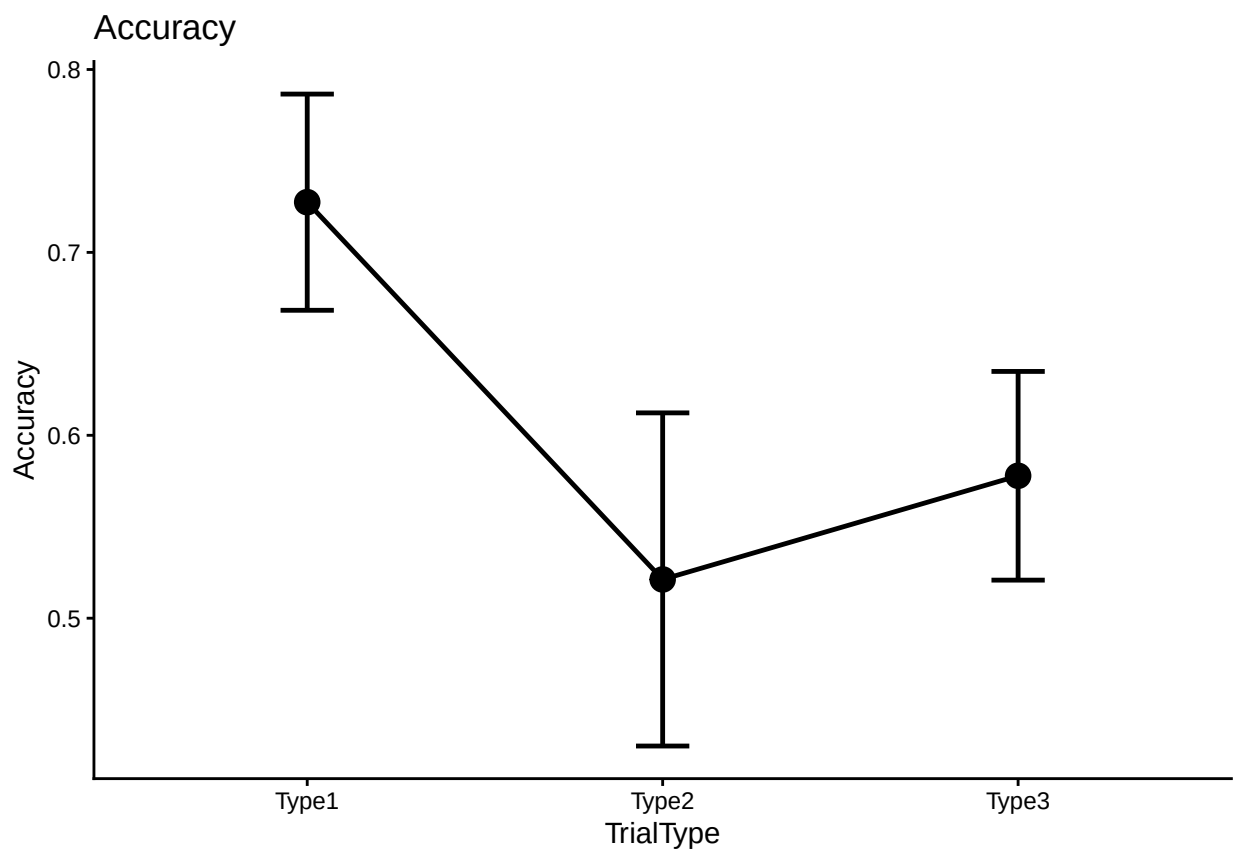
```

```
# determine the mean accuracy for question types 1, 2 and 3
```

```
accplot2 <- all_data_no_outliers %>%
  group_by(ParticipantId, TrialType) %>%
  summarize(overallmean = mean(Correct_Avg)) %>%
  group_by(TrialType) %>%
  summarize(overall_trialtype_mean = mean(overallmean),
            se = std.error(overallmean),
            n = n(),
            CI = qt(0.975,df=n-1)*se)
```

```
## `summarise()` has grouped output by 'ParticipantId'. You can override using the `.groups`
## argument.
```

```
ggplot(accplot2, aes(TrialType,
                     overall_trialtype_mean,
                     group = 1,
                     ymin = overall_trialtype_mean - CI,
                     ymax = overall_trialtype_mean + CI)) +
  theme_classic() +
  geom_point(size = 4) +
  geom_errorbar(width = .15, linewidth = 0.85) +
  geom_line(linewidth = 0.85) +
  labs(y = "Accuracy", title = "Accuracy")
```



```
kable(accplot2)
```

| TrialType | overall_trialtype_mean | se        | n   | CI        |
|-----------|------------------------|-----------|-----|-----------|
| Type1     | 0.7274590              | 0.0298540 | 122 | 0.0591038 |
| Type2     | 0.5211864              | 0.0459870 | 118 | 0.0910749 |
| Type3     | 0.5778689              | 0.0288189 | 122 | 0.0570546 |